Executable-Aware Select-and-Compress Distillation for Reliable Chain-of-Thought

Anonymous Author(s)
Affiliation
Address
email

Abstract

Noisy teacher rationales remain a principal bottleneck for chain-of-thought (CoT) distillation: students inherit arithmetic slips, unsupported statements, and redundant detours that degrade faithfulness, robustness, and efficiency. We introduce EXACT, an executable-aware select-and-compress distillation pipeline that inserts quality control before learning. EXACT verifies teacher steps with task-appropriate modules (symbolic execution for math; retrieval-augmented natural language inference for commonsense), extracts a minimal verified core via backward dependency tracing and a token-cost set cover, trains with a contrastive faithfulness objective that rewards verified core steps and penalizes rejected steps, and applies a curriculum that progressively masks intermediate verified steps to cultivate skip-step competence. The pipeline is plug-and-play: it preprocesses teacher traces and adds lightweight auxiliary heads to standard decoder-only students. We design three evaluations spanning math (GSM8K $\rightarrow$ GSM-Hard), commonsense (CommonsenseQA, HellaSwag), and cross-lingual reasoning (MGSM), and report a smoke-test execution that exercises the full pipeline. In this toy run with a tiny non-instruction-tuned student and minimal data, all methods - EXACT and baselines - performed at chance. We analyze failure modes (capacity, data, and training budget) and provide a reproducible path to scale-appropriate evaluation. Our contribution is a principled, verification-first CoT distillation framework that clarifies where verifiers, minimal cores, contrastive objectives, and compression curricula enter, enabling robust gains when deployed at realistic scales.

1 Introduction

Chain-of-thought (CoT) prompting elicits intermediate reasoning that substantially improves arithmetic and symbolic performance in large language models Wei et al. [2022]. Distilling such capabilities into smaller students is desirable for efficiency and deployability, but current CoT distillation inherits a critical liability: teacher rationales contain noisy or incorrect steps. Students trained to imitate whole traces reproduce arithmetic slips, unsupported factual claims, and needless detours, which harms faithfulness, robustness to distribution shift, and runtime efficiency through longer outputs. Recent strategies reweight key tokens and distill progressively within rationales Feng et al. [2024], scale synthetic reasoning corpora Toshniwal et al. [2024], or specialize compact models for multi-step math reasoning Fu et al. [2023], yet they generally assume that teacher traces are fully correct and useful. In practice, empirical audits reveal nontrivial error rates even in frontier models; without explicit quality control, students learn both answers and mistakes, amplifying error propagation and brittleness (e.g., on GSM-Hard and adversarial paraphrases).

Verification offers a remedy. For math, verifiers based on symbolic execution or lightweight program execution can check step-level assignments and equalities and have improved solution selection Cobbe et al. [2021], Andor et al. [2019]. For non-executable commonsense and trivia, retrieval-augmented verification combined with natural language inference (NLI) can determine whether a step is entailed or contradicted Devlin et al. [2018], Camburu et al. [2018], Bostrom et al. [2021]. Distilling verified reasoning has also proven effective in multimodal settings via program execution and verification prior to training Hu et al. [2023]. However, a general-purpose, verification-aware CoT distillation pipeline that automatically selects, repairs, and compresses teacher reasoning before student training is still missing.

We propose EXACT (Executable-Aware Select-and-Compress Distillation). EXACT interposes a step-level verification layer between teacher generation and student training, labels steps as verified, rejected, or unverifiable, and extracts the smallest verified core that still suffices to reproduce or justify the final answer. Distillation then treats verified and rejected content asymmetrically via a contrastive faithfulness objective, while a curriculum progressively hides intermediate verified steps to encourage concise, flexible reasoning. EXACT is model-agnostic and plug-and-play: it adds a preprocessing pipeline and lightweight auxiliary heads to standard decoder-only students, without modifying teacher usage beyond encouraging structured, numbered steps.

Our contributions are:

- **Verification-aware framework**: A verification-aware CoT distillation framework that couples symbolic execution for math with retrieval+NLI verification for commonsense to produce token-level verified/rejected labels for teacher steps Cobbe et al. [2021], Devlin et al. [2018], Camburu et al. [2018].
- **Minimal core extraction**: Minimal core extraction via backward dependency tracing and a token-cost set cover with executor validation, yielding compressed, guaranteed-supporting rationales.
- **Contrastive faithfulness objective**: A contrastive faithfulness objective that rewards verified core steps and penalizes rejected steps, reducing error propagation beyond keypoint weighting alone Feng et al. [2024].
- **Compression curriculum**: A curriculum-progressive compression schedule that begins with full verified rationales and gradually masks intermediate steps to train skip-step competence Liu et al. [2024].
- **End-to-end pipeline and metrics**: An end-to-end, reproducible pipeline spanning math, commonsense, and cross-lingual settings (GSM8K $\rightarrow$ GSM-Hard, CommonsenseQA, HellaSwag, MGSM), with metrics covering answer accuracy, faithfulness to verified cores, error propagation, rationale length, latency, hallucination, robustness, and multilingual transfer Cobbe et al. [2021], Shi et al. [2022], Srivastava et al. [2022].

We also provide an executed, toy-scale smoke test. Using a tiny non-instruction-tuned student with minimal training data, all methods - EXACT and baselines - achieved near-zero accuracy with no observable gains. We analyze these negative results, attributing them to capacity, data scarcity, and short training, and outline a scale-appropriate regimen (7B students with LoRA, tens of thousands of verified traces per dataset, calibrated verifiers, and self-consistency decoding). While the smoke test does not validate our expected improvements, it demonstrates that verification, minimal core extraction, contrastive training, and curriculum masking integrate coherently in a single codebase.

Beyond CoT, our approach connects to knowledge and dataset distillation Padmanabhan et al. [2023], Chen et al. [2023], Yang et al. [2024], progressive or token-adaptive methods Dennis et al. [2023], aut [b], and broader distillation frameworks that leverage auxiliary variables and structured supervision aut [a,d,c]. EXACT contributes a verification-first perspective compatible with these advances, emphasizing quality-controlled rationales as training targets.

Looking ahead, we plan to extend executable verifiers to code and scientific domains Austin et al. [2021], Chen et al. [2021], Chevalier et al. [2024], strengthen entailment models, and explore on-the-fly self-verification at inference time using the learned error-mask head.

2 Related Work

Chain-of-thought prompting and distillation. CoT prompting improves reasoning in sufficiently large models Wei et al. [2022]. Auto-CoT automates exemplar generation and mitigates errors through diversity Zhang et al. [2022], complementing concerns about overfitting on math benchmarks Zhang et al. [2024]. KPOD reweights key tokens and distills progressively from later to earlier steps Feng et al. [2024]. Unlike these methods, EXACT does not assume that all teacher steps are useful or correct: it explicitly verifies and filters steps prior to training and adds negative supervision against rejected content, aiming to reduce error propagation rather than only prioritizing key tokens.

Verification and program execution. Training verifiers for math solution selection shows clear gains on GSM8K Cobbe et al. [2021]; shallow program execution augments numerical reasoning in reading comprehension Andor et al. [2019]. Visual Program Distillation samples programs, executes and verifies them, then distills verified language descriptions into a compact vision-language model Hu et al. [2023]. EXACT follows a similar verify-then-distill principle, but targets textual CoT: verify steps, select a minimal verified support, then train the student to imitate the core while avoiding rejected content.

Knowledge-augmented and retrieval-grounded reasoning. KARD augments rationales with external knowledge and improves small models on knowledge-intensive tasks Kang et al. [2023]. In contrast, EXACT uses retrieval+NLI as a verifier rather than a generator: steps are labeled entailed, contradicted, or unverifiable using strong NLI models Devlin et al. [2018], Camburu et al. [2018], Bostrom et al. [2021]. This inversion enables explicit supervision for what not to copy, improving faithfulness.

Specialization and compression. Specializing models no larger than 11B parameters toward multi-step reasoning demonstrates notable gains Fu et al. [2023]. Dataset distillation explores compressing training dynamics Chen et al. [2023], Yang et al. [2024], and progressive ensembles trade accuracy for cost at inference Dennis et al. [2023]. EXACT's curriculum-progressive compression trains students to reason with fewer steps while maintaining correctness, related to emerging step-skipping competence Liu et al. [2024].

Benchmarks and multilingual reasoning. GSM8K is the de facto arithmetic benchmark, often paired with verifier-based selection Cobbe et al. [2021]. BIG-bench explores capability shifts with scale Srivastava et al. [2022]. MGSM demonstrates multilingual CoT abilities and cross-lingual transfer Shi et al. [2022]. EXACT targets GSM8K and GSM-Hard for math, CommonsenseQA and HellaSwag for commonsense, and MGSM for multilingual evaluation.

General distillation and efficiency. Context and knowledge distillation can propagate structured knowledge into smaller models Padmanabhan et al. [2023]. Token-adaptive compute aut [b], auxiliary-variable distillation aut [a], feature-correlation distillation aut [d], and scaffolding aut [c] are complementary. EXACT adds a verified, contrastive target for rationale tokens, orthogonal to distribution- or activation-matching.

Rationales and robustness. Generating rationales can help robustness but may be sensitive to positional or lexical artifacts Chen et al. [2022]. EXACT's unlikelihood penalty on contradicted steps is designed to suppress recurrent teacher errors and reduce hallucination, especially under adversarial paraphrases or retrieval degradation.

3 Background

Problem setting. A teacher model produces a rationale as a sequence of steps followed by a final answer. Standard CoT distillation trains a student to imitate the entire rationale and answer. In practice, teacher traces contain errors and redundancies. Naive imitation propagates these artifacts into the student, degrading accuracy, faithfulness to valid reasoning, robustness under shift, and efficiency due to unnecessarily long reasoning.

Verification resources. For math, symbolic execution and sandboxed interpreters can check step-level assignments and equalities while maintaining a state over named quantities Cobbe et al. [2021], Andor et al. [2019]. For commonsense and trivia, retrieval-augmented verification uses dense retrieval over a large corpus and an NLI classifier to label steps as entailed or contradicted Devlin et al. [2018], Camburu et al. [2018], Bostrom et al. [2021]. We adopt high-precision decision thresholds to avoid false positives, accepting reduced recall.

Core rationale extraction. Given step-level labels, we seek the smallest subset of verified steps that suffices to recompute or justify the final answer. We pose this as token-cost set cover: each verified step covers symbols or facts it introduces; the objective minimizes total token length subject to covering the required symbols or decisive facts. For math, required symbols include numerals or named quantities appearing in the answer line; for commonsense, they include explicit references to the chosen option or decisive eliminations. We preserve original order and validate selected cores by executing them (math) or aggregating entailment (commonsense).

Distillation objective. Training combines positive and negative supervision. A positive term maximizes the likelihood of verified core tokens and the final answer. A negative term applies unlikelihood to tokens from rejected steps, discouraging the student from emitting erroneous patterns. An auxiliary token classification head predicts keep versus reject labels on teacher tokens to inject additional gradients. This differs from keypoint reweighting Feng et al. [2024] by explicitly supervising the student away from incorrect teacher behaviors.

Curriculum and compression. A curriculum exposes full verified rationales initially, then progressively masks intermediate verified steps while preserving those closest to the answer and those covering required symbols. This induces skip-step competence, consistent with recent evidence that models can learn to shorten reasoning without sacrificing accuracy Liu et al. [2024].

Evaluation metrics. We track normalized exact match (answers), faithfulness F1 between generated rationales and verified cores via alignment, error-propagation rate measured as overlap with teacher-rejected tokens, rationale length and decoding throughput, hallucination rates for non-executable steps, robustness to paraphrasing or retrieval degradation, and multilingual transfer on MGSM Cobbe et al. [2021], Shi et al. [2022], Srivastava et al. [2022].

Assumptions. We prioritize precision in verification labels. Selected cores are validated for executability or support. Retrieval+NLI verification is cached and batched for efficiency. These choices reflect a verification-first approach compatible with broader distillation mechanisms aut [a,d,b,c].

4 Method

EXACT: Executable-Aware Select-and-Compress Distillation.

4.1 Verification of Teacher Steps

We segment teacher rationales into steps using numbering or line breaks. For math, we maintain a symbolic state over variables and parse assignments and equalities. Right-hand sides are evaluated and substituted; numeric agreement is checked under a tolerance. A step is labeled verified if at least one assignment or equality is consistent with the maintained state; rejected if explicit contradictions are detected; otherwise unverifiable. For commonsense and trivia, we build a retrieval index over Wikipedia, retrieve top passages using a dense retriever, and compute entailment and contradiction with a strong NLI model Devlin et al. [2018], Camburu et al. [2018]. A step is verified if any retrieved passage entails it above a calibrated threshold and contradiction is lower; rejected if contradiction exceeds threshold; else unverifiable. Thresholds are calibrated for high precision.

4.2 Minimal Core Extraction

We identify required symbols or facts needed to support the final answer. For math, required symbols include numerals and named quantities present in the answer line; for commonsense, required facts include references to the chosen option and decisive eliminations. We solve a minimum-cost set cover where each verified step covers the symbols or facts it introduces, with cost proportional to step token length. We use an integer linear program with a greedy fallback. We validate the selected core by re-executing steps (math) or checking that entailment support for the chosen answer exceeds alternatives (commonsense). If validation fails, we add nearby verified steps iteratively until success.

4.3 Contrastive Faithfulness Objective

We fine-tune a decoder-only student with parameter-efficient adapters. The objective maximizes likelihood on the final answer span and on tokens within verified core steps, optionally up-weighting core tokens. We add an unlikelihood penalty on tokens within rejected steps, discouraging the model from emitting those sequences. A small auxiliary token classification head on top of hidden states predicts keep versus reject labels for teacher tokens during training; it is not used at inference.

Relative to keypoint reweighting Feng et al. [2024], this introduces explicit negative supervision against teacher errors.

4.4 Curriculum-Progressive Compression

Across epochs, we gradually mask intermediate verified steps, keeping those closest to the final answer and those covering required symbols. Masking ratios increase (for example, 0%, then 50%, then 80%). This schedule trains students to infer elided steps and to produce shorter, accurate rationales, aligning with observations on step-skipping competence Liu et al. [2024].

4.5 Plug-and-Play Integration and Evaluation

EXACT is delivered as a preprocessing pipeline that augments each example with step strings, verified and rejected masks, and core indices, plus minor modifications to the student's loss and an optional auxiliary head. No changes to teacher prompts are required beyond encouraging structured, numbered steps. Evaluation follows the metrics in Background: exact match accuracy, faithfulness F1 to verified cores, error propagation, rationale length, throughput, hallucination rates for non-executable steps, and robustness measures Cobbe et al. [2021], Srivastava et al. [2022].

[H] EXACT: Verify-Select-Compress Distillation [1] **Input:** teacher traces $\mathcal{T}$; task type $\tau \in \{math, commonsense\}$; verifier modules (executor or retriever+NLI); masking schedule $\{m_e\}$; loss weights $\lambda_{core}, \lambda_{unlike}$ **Output:** fine-tuned student parameters θ each example $t \in \mathcal{T}$ Segment rationale into ordered steps $S = [s_1, \ldots, s_K]$ and final answer a $\tau = math$ Initialize symbolic state $\Sigma \leftarrow \emptyset$ step s_k Parse assignments/equalities; evaluate RHS; update/check Σ Label $y_k \in \{verified, rejected, unverifiable\}$ by numeric consistency step s_k Retrieve passages P_k; compute entailment e_k and contradiction c_k Label y_k by calibrated thresholds on (e_k, c_k) Identify required symbols/facts R from a and prompt Build set-cover instance: each verified step s_k covers $C_k \subseteq R$ with cost $len(s_k)$ Solve $\min \sum_k c_k z_k$ s.t. $\cup_{k:z_k=1} C_k = R$, $z_k \in \{0,1\}$; obtain core indices $\mathcal{K}$ Validate core: execute (math) or check entailment support (commonsense); if fail, expand $\mathcal{K}$ greedily Store training targets: tokens of a, tokens of steps in $\mathcal{K}$ (core mask), tokens of rejected steps (reject mask) epoch $e = 1, \ldots, E$ Determine masking ratio m_e; randomly mask a fraction m_e of intermediate verified non-core steps minibatch B Compute loss: $\mathcal{L} = \lambda_{core}\, CE(coretokens + a)$ $+ \lambda_{unlike}\, UL(rejectedtokens) + CE(promptandunmaskedverifiedtokens)$ Update student parameters $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ θ

5 Experimental Setup

Planned evaluations. We instantiate EXACT in three settings and persist JSONL artifacts at each stage: raw teacher generations; verified step labels with metadata; minimal cores; and training-ready targets with token masks and curriculum levels. Baselines include answer-only knowledge distillation, standard CoT distillation, and specialized CoT distillation such as KPOD for comparative context Feng et al. [2024]. Datasets include GSM8K for training, GSM8K test and GSM-Hard for math robustness, CommonsenseQA and HellaSwag for commonsense, and MGSM for multilingual transfer Cobbe et al. [2021], Shi et al. [2022].

Experiment 1 (Math). Objective: assess answer accuracy, faithfulness, error propagation, length, latency, and robustness; quantify contributions of verification, core extraction, contrastive loss, and curriculum. Verification uses symbolic execution to label steps. Minimal core is solved as token-cost set cover with executor validation. Students are fine-tuned with LoRA adapters; the curriculum masks intermediate verified steps across epochs. Metrics: normalized exact match; faithfulness F1; error-propagation rate; average generated length and decoding throughput; robustness on GSM-Hard and adversarial paraphrases.

Experiment 2 (Commonsense/trivia). Objective: reduce hallucination and improve accuracy via retrieval-grounded verification. Retrieval plus NLI labels steps as entailed versus contradicted; minimal core selects the smallest entailed subset supporting the final choice. Losses mirror Experiment 1, with higher unlikelihood weight for contradicted steps. Metrics: accuracy, hallucination rates, faithfulness, length and latency, and robustness to retrieval degradation.

Experiment 3 (Cross-lingual MGSM). Objective: test whether executable-aware core reasoning learned in English transfers to other languages without multilingual supervision. Evaluate two inference modes: standard and compressed ("use minimal steps"), leveraging the curriculum. A language-agnostic faithfulness proxy verifies numeric equalities in generated rationales with symbolic execution. Metrics: per-language accuracy, output length, and numeric verification rate Shi et al. [2022].

Executed smoke test. We conducted an end-to-end, toy-scale execution to validate the pipeline under constrained compute. Hardware: a single CUDA device reported as Tesla T4. Student: a tiny GPT-2-like model without instruction tuning and without LoRA, trained for 2 epochs with small batches. Teacher traces were synthetic and few in number. EXACT losses included weighted cross-entropy on verified core tokens and unlikelihood on rejected tokens; the curriculum hid 0% then 50% of intermediate verified steps. Baselines: standard CoT distillation (all steps treated as core; no contrastive term) and answer-only distillation. The full verification and minimal-core components were active: SymPy-based execution for math; retrieval plus NLI for commonsense. Evaluation used greedy decoding with a fixed maximum number of new tokens. We recorded answer accuracy, faithfulness F1 to the verified core via alignment, error-propagation rate, average generated length, decoding throughput, and hallucination rates for commonsense.

Notes on scope. This smoke test is not a substitute for the planned scale (instruction-tuned 7B students with LoRA trained on tens of thousands of verified rationales). It verifies that verification, core extraction, contrastive training, curriculum masking, and metric computation integrate coherently in a single codebase, while highlighting the need for scale to observe the hypothesized benefits supported by prior work Cobbe et al. [2021], Feng et al. [2024], Kang et al. [2023].

6 Results

We report the executed toy-scale run that exercised the EXACT pipeline and two baselines. All numbers below come directly from the logs; no other results are claimed.

Experiment 1 (Math, GSM-like). EXACT: accuracy 0.000; faithfulness F1 0.000; error-propagation rate 0.024; average generated length 164 tokens; decoding throughput approximately 533.3 tokens per second. Standard CoT distillation: accuracy 0.000; faithfulness F1 0.000; error-propagation rate 0.024; length 164; approximately 564.1 tokens per second. Answer-only distillation: accuracy 0.000; faithfulness F1 0.000; error-propagation rate 0.024; length 164; approximately 572.8 tokens per second. Training losses remained around 10.82 across epochs for all methods. The accuracy and length comparisons for this experiment are summarized in Figure 1 and Figure 6; error propagation and faithfulness appear in Figure 9 and Figure 10; inference latency is shown in Figure 12.

Experiment 2 (Commonsense). EXACT: accuracy 0.000; hallucination rate 0.875; average length 183; throughput approximately 917.5 tokens per second. Standard CoT distillation: accuracy 0.000; hallucination 0.875; length 183; approximately 914.4 tokens per second. Answer-only distillation: accuracy 0.000; hallucination 0.875; length 183; approximately 935.9 tokens per second. Training losses were approximately 10.82 across epochs. The accuracy and length comparisons appear in Figure 2 and Figure 7; hallucination rates in Figure 11; confusion matrices for EXACT and the baseline are provided in Figure 5 and Figure 4, respectively.

Experiment 3 (Cross-lingual MGSM-style). For English, Spanish, French, and German: accuracy 0.000 in both standard and compressed modes; average lengths roughly 161-175 tokens in standard mode and 166-180 in compressed mode; numeric faithfulness proxy approximately 0.00 in all evaluated languages. The accuracy and length summaries are shown in Figure 3 and Figure 8.

Interpretation and limitations. Under severe data and capacity constraints, the student failed to learn task formatting or to imitate verified core steps, yielding zero accuracy and zero faithfulness across methods. Error propagation rates were uniformly low not due to deliberate avoidance of errors but because generations had little overlap with teacher tokens. Commonsense hallucination remained high, reflecting ungrounded generations relative to the retrieval verifier. These negative results should be interpreted as pipeline validation rather than scientific evaluation of EXACT's hypotheses, which require scale-appropriate models, sufficient training data, and longer schedules. The planned regime - 7B students with LoRA, approximately 30k verified traces per dataset, calibrated NLI thresholds, and self-consistency decoding - is consistent with prior findings that verification and structured distillation improve reasoning Cobbe et al. [2021], Feng et al. [2024], Kang et al. [2023].

Recommendations for meaningful evaluation. Use instruction-tuned 7B students with LoRA; collect multiple teacher rationales per item to raise verification recall; maintain high-precision thresholds for the NLI verifier Devlin et al. [2018], Camburu et al. [2018]; enforce answer formatting with constrained decoding; evaluate with self-consistency; and report accuracy, faithfulness, error propagation, length, latency, and hallucination. Run ablations removing verification, the contrastive penalty, and the curriculum; vary compression ratios; and compare to keypoint/progressive baselines Feng et al. [2024].

7 Conclusion

We presented EXACT, a verification-aware, select-and-compress approach to chain-of-thought distillation. EXACT integrates step-level verification (symbolic execution for math; retrieval plus NLI for commonsense), minimal core extraction via token-cost set cover and executor validation, a contrastive faithfulness objective that rewards verified steps and penalizes rejected ones, and a curriculum that progressively masks intermediate steps to induce concise, skip-step reasoning. The pipeline is plug-and-play, coupling principled preprocessing with lightweight student-side changes.

Our end-to-end, toy-scale execution confirmed that the pipeline components integrate and run but yielded chance performance across all methods. We diagnose the causes as student capacity limits, data scarcity, and insufficient training budget. The path forward is clear: scale to instruction-tuned 7B students with LoRA; collect tens of thousands of verified rationales per dataset; calibrate verifiers for high precision; enforce answer formatting and decode with self-consistency; and evaluate with comprehensive metrics and ablations. Prior work suggests that, under these conditions, verification and structured distillation can yield meaningful gains in accuracy, faithfulness, robustness, and efficiency Cobbe et al. [2021], Feng et al. [2024], Kang et al. [2023].

Future directions include extending executable verifiers to code and scientific domains Austin et al. [2021], Chen et al. [2021], Chevalier et al. [2024], strengthening entailment models and retrieval robustness, and exploring on-the-fly self-verification at inference time via the learned error-mask head. EXACT offers a principled foundation for quality-controlled CoT distillation and a practical path toward democratizing reliable reasoning in compact models.

References

Knowledge distillation with auxiliary variable. a.

D-llm: A token adaptive computing resource allocation strategy for large language models. b.

Scaffolding a student to instill knowledge. c.

Towards efficient pre-trained language model via feature correlation distillation. d.

Daniel Andor, Luheng He, Kenton Lee, and Emily Pitler. Giving bert a calculator: Finding operations and arguments with reading comprehension. 2019.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program synthesis with large language models. 2021.

Kaj Bostrom, Xinyu Zhao, Swarat Chaudhuri, and Greg Durrett. Flexible generation of natural language deductions. 2021.

Oana-Maria Camburu, Tim Rocktäschel, Thomas Lukasiewicz, and Phil Blunsom. e-snli: Natural language inference with natural language explanations. 2018.

Howard Chen, Jacqueline He, Karthik Narasimhan, and Danqi Chen. Can rationalization improve robustness? 2022.

[width=0.7] images/accuracy_exp1.pdf

Figure 1: Accuracy comparison for Experiment 1

[width=0.7] images/accuracy_exp2.pdf

Figure 2: Accuracy comparison for Experiment 2

[width=0.7] images/accuracy_exp3.pdf

Figure 3: Accuracy comparison for Experiment 3

[width=0.7] images/confusion_matrix_exp2_baseline.pdf

Figure 4: Confusion matrix for commonsense baseline

[width=0.7] images/confusion_matrix_exp2_exact.pdf

Figure 5: Confusion matrix for commonsense EXACT

[width=0.7] images/cot_length_exp1.pdf

Figure 6: Rationale length for Experiment 1

[width=0.7] images/cot_length_exp2.pdf

Figure 7: Rationale length for Experiment 2

[width=0.7] images/cot_length_exp3.pdf

Figure 8: Rationale length for Experiment 3

[width=0.7] images/epr_exp1.pdf

Figure 9: Error propagation rate for Experiment 1

[width=0.7] images/faithfulness_exp1.pdf

Figure 10: Faithfulness F1 for Experiment 1

[width=0.7] images/hallucination_exp2.pdf

Figure 11: Hallucination rates for Experiment 2

[width=0.7] images/inference_latency_exp1.pdf

Figure 12: Inference latency for Experiment 1

[width=0.7] images/training_loss_exp1_baseline.pdf

Figure 13: Training loss for Experiment 1 baseline

[width=0.7] images/training_loss_exp2_exact.pdf

Figure 14: Training loss for Experiment 2 EXACT

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. 2021.

Xuxi Chen, Yu Yang, Zhangyang Wang, and Baharan Mirzasoleiman. Data distillation can be like vodka: Distilling more times for better quality. 2023.

Alexis Chevalier, Jiayi Geng, Alexander Wettig, Howard Chen, Sebastian Mizera, Toni Annala, Max Jameson Aragon, Arturo Rodríguez Fanlo, Simon Frieder, Simon Machado, Akshara Prabhakar, Ellie Thieu, Jiachen T. Wang, Zirui Wang, Xindi Wu, Mengzhou Xia, Wenhan Xia, Jiatong Yu, Jun-Jie Zhu, Zhiyong Jason Ren, Sanjeev Arora, and Danqi Chen. Language models as science tutors. 2024.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. 2021.

Don Kurian Dennis, Abhishek Shetty, Anish Sevekari, Kazuhito Koishida, and Virginia Smith. Progressive ensemble distillation: Building ensembles for efficient inference. 2023.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. 2018.

Kaituo Feng, Changsheng Li, Xiaolu Zhang, Jun Zhou, Ye Yuan, and Guoren Wang. Keypoint-based progressive chain-of-thought distillation for llms. 2024.

Yao Fu, Hao Peng, Litu Ou, Ashish Sabharwal, and Tushar Khot. Specializing smaller language models towards multi-step reasoning. 2023.

Yushi Hu, Otilia Stretcu, Chun-Ta Lu, Krishnamurthy Viswanathan, Kenji Hata, Enming Luo, Ranjay Krishna, and Ariel Fuxman. Visual program distillation: Distilling tools and programmatic reasoning into vision-language models. 2023.

Minki Kang, Seanie Lee, Jinheon Baek, Kenji Kawaguchi, and Sung Ju Hwang. Knowledge-augmented reasoning distillation for small language models in knowledge-intensive tasks. 2023.

Tengxiao Liu, Qipeng Guo, Xiangkun Hu, Cheng Jiayang, Yue Zhang, Xipeng Qiu, and Zheng Zhang. Can language models learn to skip steps? 2024.

Shankar Padmanabhan, Yasumasa Onoe, Michael J. Q. Zhang, Greg Durrett, and Eunsol Choi. Propagating knowledge updates to lms through distillation. 2023.

Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. Language models are multilingual chain-of-thought reasoners. 2022.

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, Agnieszka Kluska, Aitor Lewkowycz, Akshat Agarwal, Alethea Power, Alex Ray, Alex Warstadt, Alexander W. Kocurek, Ali Safaya, Ali Tazarv, Alice Xiang, Alicia Parrish, Allen Nie, Aman Hussain, Amanda Askell, Amanda Dsouza, Ambrose Slone, Ameet Rahane, Anantharaman S. Iyer, Anders Andreassen, Andrea Madotto, Andrea Santilli, Andreas Stuhlmüller, Andrew Dai, Andrew La, Andrew Lampinen, Andy Zou, Angela Jiang, Angelica Chen, Anh Vuong, Animesh Gupta, Anna Gottardi, Antonio Norelli, Anu Venkatesh, Arash Gholamidavoodi, Arfa Tabassum, Arul Menezes, Arun Kirubarajan, Asher Mullokandov, Ashish Sabharwal, Austin Herrick, Avia Efrat, Aykut

Erdem, Ayla Karakaş, B. Ryan Roberts, Bao Sheng Loe, Barret Zoph, Bartłomiej Bojanowski, Batuhan Özyurt, Behnam Hedayatnia, Behnam Neyshabur, Benjamin Inden, Benno Stein, Berk Ekmekci, Bill Yuchen Lin, Blake Howald, Bryan Orinion, Cameron Diao, Cameron Dour, Catherine Stinson, Cedrick Argueta, César Ferri Ramírez, Chandan Singh, Charles Rathkopf, Chenlin Meng, Chitta Baral, Chiyu Wu, Chris Callison-Burch, Chris Waites, Christian Voigt, Christopher D. Manning, Christopher Potts, Cindy Ramirez, Clara E. Rivera, Clemencia Siro, Colin Raffel, Courtney Ashcraft, Cristina Garbacea, Damien Sileo, Dan Garrette, Dan Hendrycks, Dan Kilman, Dan Roth, Daniel Freeman, Daniel Khashabi, Daniel Levy, Daniel Moseguí González, Danielle Perszyk, Danny Hernandez, Danqi Chen, Daphne Ippolito, Dar Gilboa, David Dohan, David Drakard, David Jurgens, Debajyoti Datta, Deep Ganguli, Denis Emelin, Denis Kleyko, Deniz Yuret, Derek Chen, Derek Tam, Dieuwke Hupkes, Diganta Misra, Dilyar Buzan, Dimitri Coelho Mollo, Diyi Yang, Dong-Ho Lee, Dylan Schrader, Ekaterina Shutova, Ekin Dogus Cubuk, Elad Segal, Eleanor Hagerman, Elizabeth Barnes, Elizabeth Donoway, Ellie Pavlick, Emanuele Rodola, Emma Lam, Eric Chu, Eric Tang, Erkut Erdem, Ernie Chang, Ethan A. Chi, Ethan Dyer, Ethan Jerzak, Ethan Kim, Eunice Engefu Manyasi, Evgenii Zheltonozhskii, Fanyue Xia, Fatemeh Siar, Fernando Martínez-Plumed, Francesca Happé, Francois Chollet, Frieda Rong, Gaurav Mishra, Genta Indra Winata, Gerard de Melo, Germán Kruszewski, Giambattista Parascandolo, Giorgio Mariani, Gloria Wang, Gonzalo Jaimovitch-López, Gregor Betz, Guy Gur-Ari, Hana Galijasevic, Hannah Kim, Hannah Rashkin, Hannaneh Hajishirzi, Harsh Mehta, Hayden Bogar, Henry Shevlin, Hinrich Schütze, Hiromu Yakura, Hongming Zhang, Hugh Mee Wong, Ian Ng, Isaac Noble, Jaap Jumelet, Jack Geissinger, Jackson Kernion, Jacob Hilton, Jaehoon Lee, Jaime Fernández Fisac, James B. Simon, James Koppel, James Zheng, James Zou, Jan Kocoń, Jana Thompson, Janelle Wingfield, Jared Kaplan, Jarema Radom, Jascha Sohl-Dickstein, Jason Phang, Jason Wei, Jason Yosinski, Jekaterina Novikova, Jelle Bosscher, Jennifer Marsh, Jeremy Kim, Jeroen Taal, Jesse Engel, Jesujoba Alabi, Jiacheng Xu, Jiaming Song, Jillian Tang, Joan Waweru, John Burden, John Miller, John U. Balis, Jonathan Batchelder, Jonathan Berant, Jörg Frohberg, Jos Rozen, Jose Hernandez-Orallo, Joseph Boudeman, Joseph Guerr, Joseph Jones, Joshua B. Tenenbaum, Joshua S. Rule, Joyce Chua, Kamil Kanclerz, Karen Livescu, Karl Krauth, Karthik Gopalakrishnan, Katerina Ignatyeva, Katja Markert, Kaustubh D. Dhole, Kevin Gimpel, Kevin Omondi, Kory Mathewson, Kristen Chiafullo, Ksenia Shkaruta, Kumar Shridhar, Kyle McDonell, Kyle Richardson, Laria Reynolds, Leo Gao, Li Zhang, Liam Dugan, Lianhui Qin, Lidia Contreras-Ochando, Louis-Philippe Morency, Luca Moschella, Lucas Lam, Lucy Noble, Ludwig Schmidt, Luheng He, Luis Oliveros Colón, Luke Metz, Lütfi Kerem Şenel, Maarten Bosma, Maarten Sap, Maartje ter Hoeve, Maheen Farooqi, Manaal Faruqui, Mantas Mazeika, Marco Baturan, Marco Marelli, Marco Maru, Maria Jose Ramírez Quintana, Marie Tolkiehn, Mario Giulianelli, Martha Lewis, Martin Potthast, Matthew L. Leavitt, Matthias Hagen, Mátyás Schubert, Medina Orduna Baitemirova, Melody Arnaud, Melvin McElrath, Michael A. Yee, Michael Cohen, Michael Gu, Michael Ivanitskiy, Michael Starritt, Michael Strube, Michał Swędrowski, Michele Bevilacqua, Michihiro Yasunaga, Mihir Kale, Mike Cain, Mimee Xu, Mirac Suzgun, Mitch Walker, Mo Tiwari, Mohit Bansal, Moin Aminnaseri, Mor Geva, Mozhdeh Gheini, Mukund Varma T, Nanyun Peng, Nathan A. Chi, Nayeon Lee, Neta Gur-Ari Krakover, Nicholas Cameron, Nicholas Roberts, Nick Doiron, Nicole Martinez, Nikita Nangia, Niklas Deckers, Niklas Muennighoff, Nitish Shirish Keskar, Niveditha S. Iyer, Noah Constant, Noah Fiedel, Nuan Wen, Oliver Zhang, Omar Agha, Omar Elbaghdadi, Omer Levy, Owain Evans, Pablo Antonio Moreno Casares, Parth Doshi, Pascale Fung, Paul Pu Liang, Paul Vicol, Pegah Alipoormolabashi, Peiyuan Liao, Percy Liang, Peter Chang, Peter Eckersley, Phu Mon Htut, Pinyu Hwang, Piotr Miłkowski, Piyush Patil, Pouya Pezeshkpour, Priti Oli, Qiaozhu Mei, Qing Lyu, Qinlang Chen, Rabin Banjade, Rachel Etta Rudolph, Raefer Gabriel, Rahel Habacker, Ramon Risco, Raphaël Millière, Rhythm Garg, Richard Barnes, Rif A. Saurous, Riku Arakawa, Robbe Raymaekers, Robert Frank, Rohan Sikand, Roman Novak, Roman Sitelew, Ronan LeBras, Rosanne Liu, Rowan Jacobs, Rui Zhang, Ruslan Salakhutdinov, Ryan Chi, Ryan Lee, Ryan Stovall, Ryan Teehan, Rylan Yang, Sahib Singh, Saif M. Mohammad, Sajant Anand, Sam Dillavou, Sam Shleifer, Sam Wiseman, Samuel Gruetter, Samuel R. Bowman, Samuel S. Schoenholz, Sanghyun Han, Sanjeev Kwatra, Sarah A. Rous, Sarik Ghazarian, Sayan Ghosh, Sean Casey, Sebastian Bischoff, Sebastian Gehrmann, Sebastian Schuster, Sepideh Sadeghi, Shadi Hamdan, Sharon Zhou, Shashank Srivastava, Sherry Shi, Shikhar Singh, Shima Asaadi, Shixiang Shane Gu, Shubh Pachchigar, Shubham Toshniwal, Shyam Upadhyay, Shyamolima, Debnath, Siamak Shakeri, Simon Thormeyer, Simone Melzi, Siva Reddy, Sneha Priscilla Makini, Soo-Hwan Lee, Spencer Torene, Sriharsha Hatwar, Stanislas Dehaene, Stefan Divic, Stefano Ermon, Stella Biderman, Stephanie Lin, Stephen Prasad, Steven T. Piantadosi, Stuart M. Shieber, Summer Misherghi, Svetlana Kiritchenko, Swaroop Mishra, Tal Linzen, Tal Schuster, Tao Li, Tao Yu, Tariq Ali, Tatsu Hashimoto, Te-Lin Wu, Théo Desbordes, Theodore Rothschild, Thomas Phan, Tianle Wang, Tiberius Nkinyili, Timo Schick, Timofei Kornev, Titus Tunduny, Tobias Gerstenberg, Trenton Chang, Trishala Neeraj, Tushar Khot, Tyler Shultz, Uri Shaham, Vedant Misra, Vera Demberg, Victoria Nyamai, Vikas Raunak, Vinay Ramasesh, Vinay Uday Prabhu, Vishakh Padmakumar, Vivek Srikumar, William Fedus, William Saunders, William Zhang, Wout Vossen, Xiang Ren, Xiaoyu Tong, Xinran Zhao, Xinyi Wu, Xudong Shen, Yadollah Yaghoobzadeh, Yair Lakretz, Yangqiu Song, Yasaman Bahri, Yejin Choi, Yichi Yang, Yiding Hao, Yifu Chen, Yonatan Belinkov, Yu Hou, Yufang Hou, Yuntao Bai, Zachary Seid, Zhuoye Zhao, Zijian Wang, Zijie J. Wang, Zirui Wang, and Ziyi Wu. Beyond the imitation game: Measuring and extrapolating the capabilities of language models. *Transactions on Machine Learning Research, May/2022, https://openreview.net/forum?id=uyTL5Bvosj*, 2022.

Shubham Toshniwal, Ivan Moshkov, Sean Narenthiran, Daria Gitman, Fei Jia, and Igor Gitman. Openmathinstruct-1: A 1.8 million math instruction tuning dataset. 2024.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. 2022.

William Yang, Ye Zhu, Zhiwei Deng, and Olga Russakovsky. What is dataset distillation learning? 2024.

Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, Will Song, Tiffany Zhao, Pranav Raja, Charlotte Zhuang, Dylan Slack, Qin Lyu, Sean Hendryx, Russell Kaplan, Michele Lunati, and Summer Yue. A careful examination of large language model performance on grade school arithmetic. 2024.

Zhuosheng Zhang, Aston Zhang, Mu Li, and Alex Smola. Automatic chain of thought prompting in large language models. 2022.